

I. K. Gujral Punjab Technical University
Bachelor of Computer Applications (BCA)

Internal max. marks: 40	Duration of end semester exam (ESE): 3hrs
External max. marks: 60	Elective status: Elective
Total marks: 100	

Prerequisite: -NA-

Co requisite: -NA-

Additional material required in ESE: -NA-

Course Outcomes:

CO#	Course outcomes
CO1	Define the concept of machine learning
CO2	Outline the key characteristics of machine learning algorithms
CO3	Compare the performance of different machine learning algorithms
CO4	Design solution for basic problems using machine learning algorithms
CO5	Explain the concept of reinforcement learning

Detailed Contents	Contact hours
Unit-I Introduction What is Machine Learning, Unsupervised Learning, Reinforcement Learning Machine Learning Use-Cases, Machine Learning Process Flow, Machine Learning Categories, Linear regression and Gradient descent.[CO1]	8
Unit-II Supervised Learning Classification and its use cases, Decision Tree, Algorithm for Decision Tree Induction Creating a Perfect Decision Tree, Confusion Matrix, Random Forest. What is Naïve Bayes, How Naïve Bayes works, Implementing Naïve Bayes Classifier, Support Vector Machine, Illustration how Support Vector Machine works, Hyper parameter Optimization, Grid Search Vs Random Search, Implementation of Support Vector Machine for Classification.[CO4]	12
Unit-III Clustering What is Clustering & its Use Cases, K-means Clustering, How does K-means algorithm work, C-means Clustering, Hierarchical Clustering, How Hierarchical Clustering works.[CO3]	12
Unit-IV	12

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Why Reinforcement Learning, Elements of Reinforcement Learning, Exploration vs Exploitation dilemma, Epsilon Greedy Algorithm, Markov Decision Process (MDP) Q values and V values, Q – Learning, α values.[CO5]	
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Text Books:

1. Pattern Reorganization and Machine learning by Christopher M. Bishop.
2. The elements of Statistical learning by Jerome H. Friedman, Robert Tibshirani and Trevor Hastie.
3. Introduction to Machine Learning by Ethem Alpaydin. PHI Publisher.
4. Machine Learning, A practical approach on the statistical learning theory by Rodrigo Fernandes de Mello and Moacir Antonelli Ponti.
5. Machine Learning A probabilistic perspective by Kevin P. Murphy

Course Code: UGCA1956

Course Name: Machine Learning Laboratory

Program: BCA	L: 0 T: 0 P: 4
Branch: Computer Applications	Credits: 2
Semester: 6 th	Contact hours: 4 hours per week
Internal max. marks: 70	Theory/Practical: Practical
External max. marks: 30	Duration of End Semester Exam (ESE): 3hrs
Total marks: 100	Elective status: Core

Prerequisite: Students must have the knowledge of editors like Notepad etc. and basic understanding of Scripting Language/s.

Co requisite: Knowledge of Networking, Internet, Client Server concepts, Static & Dynamic environment of the websites etc.

Additional material required in ESE:

Course Outcomes: After studying this course, students will be able to:

CO#	Course Outcomes
CO1	Differentiate between various data types.
CO2	Implement programs for various Learning algorithms.
CO3	Compare different machine learning algorithms.
CO4	Choose the right algorithm for different problems.
CO5	Apply Machine Learning algorithms to solve real world problems.

Instructions: Instructor can increase/decrease the experiments as per the requirement.

Assignments:

1.	Read the numeric data from .CSV file and use some basic operation on it.
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